

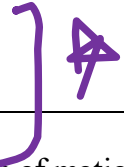
National Curriculum of Pakistan 2022-2023
Assessment Framework Physics Grade-IX (SSC-I)
Details of Content Areas/ SLOs

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
Domain: A Measurements			
[SLO: P-09-A-01] Differentiate Between physical and non-physical quantities	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-A-02] Explain with examples that physics is based on physical quantities [Including that these consist of a magnitude and a unit]	Summative	Understanding	Question(s) will be asked in annual examination.
[SLO: P-09-A-03] Differentiate between base and derived physical quantities and units. ✓	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-A-04] Apply the seven units of System International (SI) [along with their symbols and physical quantities (standard definitions of SI units are not required)]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-A-05] Analyze and express numerical data using scientific notation [In measurements and calculations]	Summative	Understanding+ Application	Question(s) will be asked in final examination.
[SLO: P-09-A-06] Analyze and express numerical data, using prefixes. [including use of their symbol to indicate decimal submultiples or multiples of both base and derived units. Specially: Pico (p), nano (n) micro (μ), milli (m), centi (c) deci (d), kilo (k), mega (M), giga (G), tera (T). This also includes: interconverting the prefixes and their symbol to indicate multiples and sub-multiples for both base and drive units.] ✓✓	Summative	Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-A-07] Differentiate between scalar and vector quantities. [A scalar has magnitude (size) only and that a vector quantity has magnitude and direction. students should be able to represent vectors graphically]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-A-08] Justify that distance, speed, time, mass, energy, and temperature are scalar quantities.	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-A-09] Justify that displacement, force; weight, velocity, acceleration, momentum, electric field strength and gravitational field strength are vector quantities. ✓✓	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-A-10] Determine, by calculation or graphically, the resultant of two vectors at right angles	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-A-11] Make reasonable estimates of physical quantities [of those that are discussed in the topics of this grade level]	Summative	Application	Question(s) will be asked in final examination.

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
[SLO: P-09-A-12] Justify and illustrate the use of common lab instruments to measure length [including how to measure a variety of lengths with appropriate precision using tapes, rulers, micrometers, and Vernier calipers (including reading the scales on analogue calipers and micrometers)]	Summative for PBA (Practical Based Assessment)	Knowledge+ Understanding+ Application	Lab work-Question(s) will be asked in PBA
[SLO: P-09-A-13] Justify and illustrate the use of measuring cylinders to measure volume [including both measurement of volume of liquid and determining the volume of a solid by displacement]	Summative for PBA (Practical Based Assessment)	Understanding+ Application	Lab work-Question(s) will be asked in PBA
[SLO: P-09-A-14] Justify and illustrate how to measure time intervals using lab instruments [including clocks and digital timers]	Summative for PBA (Practical Based Assessment)	Understanding+ Application	Lab work-Question(s) will be asked in PBA
[SLO: P-09-A-15] Determine an average value for an empirical reading [including small distance and for a short interval of time by measuring multiples (including the period of oscillations of a pendulum)]	Summative for PBA (Practical Based Assessment)	Understanding + Application	Lab work-Question(s) will be asked in PBA
[SLO: P-09-A-16] Round off and justify calculational estimates [Based on empirical data to an appropriate number of significant figures]	Summative for PBA (Practical Based Assessment)	Understanding + Application	Lab work-Question(s) will be asked in PBA
[SLO: P-09-A-17] Critique and analyze experiments for sources of error [including identifying sources of systematic and random error in measurements and suggesting steps to correct them]	Summative	Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-A-18] Differentiate between precision and accuracy	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-A-19] Determine the least count of a data collection instrument (analog) from its scale	Summative for PBA (Practical Based Assessment)	Understanding+ Application	Lab work-Question(s) will be asked in PBA

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
Domain: B Mechanics			
[SLO: P-09-B-01] Differentiate between different types of motion [i.e. translatory (linear, random and circular), rotatory and vibratory motions and distinguish among them]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-02] Differentiate between distance and displacement, speed and velocity	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-03] Define and Calculate Speed [using the equation speed = distance/time, $v = \Delta s / \Delta t$ (this should include an understanding of term instantaneous) speed]	Summative	Understanding+ Application	Question(s) will be asked in final examination.
[SLO: P-09-B-04] Define and Calculate average speed [average speed = (total distance travelled)/(total time taken)]	Summative	Knowledge+ Understanding+ Application	Question(s) will be asked in final examination.
[SLO: P-09-B-05] Differentiate between average and instantaneous speed	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-06] Differentiate between uniform velocity and non- uniform velocity	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-07] Define and calculate acceleration [includes deriving the units of acceleration as m/s^2 from the formula $a = \Delta v / \Delta t$ and using the formula to solve problems. This also includes knowing that deceleration is negative acceleration and using fact in calculations.]	Summative	Understanding+ Application	Question(s) will be asked in final examination.
[SLO: P-09-B-08] Differentiate between uniform acceleration and non-uniform acceleration	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-09] Sketch, plot and interpret distance— time and Speed-time graphs. This includes determining from the shape of a distance-time graph when an object is: [(a) at rest, (b) moving with constant speed, (c) accelerating, (d) decelerating, Students are Also required to know how to calculate speed from gradient of a distance-time graph. It also includes determining from the shape of a speed time graph when an object is: (a) at rest, (b) Moving with constant speed (c) moving with constant acceleration (d) moving with changing acceleration]	Summative	Understanding+ Application	Question(s) will be asked in final examination.
[SLO: P-09-B-10] Use the approximate value 9.8m/s^2 for free fall acceleration near Earth to solve problems	Summative	Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-11] Justify how the gradient of a distance Vs. time graph gives the speed	Summative	Understanding + Application	Question(s) will be asked in final examination.

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
[Without using calculus]			
[SLO: P-09-B-12] Analyze the Distance travelled in speed vs time graphs [by determining the area under the graph for cases of motion with constant speed or constant acceleration]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-13] Derive how the area beneath a speed Vs. time graph gives the distance traveled (without calculus)	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-14] Calculate acceleration from the gradient of a speed—time graph	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-15] Justify how the gradient of the speed vs time graph gives the acceleration [Without using calculus]	Summative	Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-16] State that there is a universal speed limit for any object in the universe that is approximately $3 \times 10^8 \text{ m/s}$ [Students should just be aware that this phenomenon is true; they do not need to study relativity in any depth. The purpose is that students appreciate that there is a universal speed limit.]	Summative	Application	Questions will be asked in annual examination.
[SLO: P-09-B-17] Illustrate that mass is a measure of the quantity of matter in an object	Summative	Knowledge	Questions will be asked in annual examination.
[SLO: P-09-B-18] Explain that the mass of an object resists change from its state of rest or motion (inertia)	Summative	Understanding	Questions will be asked in annual examination.
[SLO: P-09-B-19] Define and calculate weight [weight is the force exerted on an object with mass by a planet's gravity, and use $W=mg$]	Summative	Understanding+ Application	Question(s) will be asked in final examination.
[SLO: P-09-B-20] Define and calculate gravitational field strength [this includes being able to state that a gravitational field is a region in which a mass experiences a force due to gravitational attraction. Student should be able to define gravitational field strength (g) as force per unit mass. Use the equation gravitational field strength = weight/mass $g = W/m$ [and know that this is equivalent to the acceleration of free fall]]	Summative	Knowledge + Understanding+ Application	Question(s) will be asked in final examination.
[SLO: P-09-B-21] Justify and illustrate the use of electronic balances to measure mass [Understand the internal workings of the electronic balance is not required: just how to practically used instrument in appropriate situations]	Summative for PBA (Practical Based Assessment)	Understanding+ Application	Lab work-Question(s) will be asked in PBA
[SLO: P-09-B-22] Justify and illustrate the use of a force meter to measure weight	Summative for PBA (Practical Based	Understanding+ Application	Lab work-Question(s) will be asked in PBA

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
	Assessment)		
[SLO: P-09-B-23] Differentiate between contact and noncontact forces 	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-24] Differentiate between different types of forces [Including weight (gravitational force), friction, drag, air resistance, tension (elastic force), electrostatic force, magnetic force, thrust (driving force) and contact force]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-25] State that there are three fundamental forces and describe them in terms of their relative strengths [These are gravitational strong and electro weak forces. Student should know that Pakistani scientist won the prize Nobel Prize for helping prove that the weak force and electromagnetic force are actually unified]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-26] Represent the forces acting on a body using free body diagrams	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-27] State and apply Newton's first law 	Summative	Knowledge + Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-28] Identify the effect of force on velocity [It may change the velocity of an object by changing its direction of motion or its speed]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-29] Determine the resultant of two or more forces acting along the same straight line	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-30] State and apply Newton's second law in terms of acceleration 	Summative	Knowledge + Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-31] State and apply Newton's third law 	Summative	Knowledge + Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-32] Explain with examples how Newton's third law describes pairs of forces of the same type acting on different objects	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-33] State the limitations of Newton's laws of motion. [That they are not exact but provide a good approximation, unless an object is moving close to the speed of light or is small enough that quantum effects become significant (for example, in the case of high speed bodies, the theory of relativistic mechanics is used. In the case of very small objects at the subatomic level, quantum mechanics is used)]	Summative	Understanding	Question(s) will be asked in final examination.

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
[SLO: P-09-B-34] Describe and identify states of equilibrium [This include the types, conditions, and states of equilibrium and identifying examples of them in our daily life]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-35] Analyze the dissipative effect of friction [this include identifying where dissipation may occur and giving example such as rubbing hands together produces heat, asteroids that enter the Earth's atmosphere disintegrate due to high temperature generated from air resistance]	Summative	Application+ Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-36] Analyze the dynamics of an object reaching terminal velocity	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-37] Differentiate qualitatively between rolling and sliding friction [no need for coefficient of friction]	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-38] Justify methods to reduce friction.	Summative	Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-39] Define and calculate momentum	Summative	Knowledge + Understanding+ Application	Question(s) will be asked in final examination.
[SLO: P-09-B-40] Define and calculate impulse [use the equation impulse = $F\Delta t = m\Delta v$]	Summative	Knowledge+Understanding+Application	Question(s) will be asked in final examination.
[SLO: P-09-B-41] Apply the principle of the conservation of momentum to solve simple problems in one dimension	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-42] Define resultant force in terms of momentum [As the change in momentum per unit time; recall and use the equation resultant force change in momentum/time taken $F = \Delta p / \Delta t$]	Summative	Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-43] Differentiate between like and unlike Parallel forces.	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-44] Analyze problems involving turning effects of forces [Students should know that moment of a force = force x perpendicular distance from the pivot and be able to use this in simple problems and be able to give examples and applications of turning effects in real life]	Summative	Knowledge + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-45] Analyze objects in equilibrium using the principle of moments	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-46] Justify experiment to verify the principle of moments	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-47] State what is meant by center of mass and center of gravity	Summative	Knowledge+	Question(s) will be asked in final examination.

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
		Understanding	final examination.
[SLO: P-09-B-48] Describe how to determine the position of the center of gravity of a plane lamina using a plumb Line	Summative for PBA (Practical Based Assessment)	Understanding + Application	Lab work-Question(s) will be asked in PBA
[SLO: P-09-B-49] Analyze, qualitatively, the effect of the position of the center of gravity on the stability of simple objects	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-50] Propose how the stability of an object can be improved [by lowering the centre of mass and increasing the base area of the object]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-51] Illustrate the applications of stability physics in real life [such as this concept is Central to Engineering Technology such as balancing toys and racing cars]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-52] Predict qualitatively the motion of rotating bodies [describe qualitatively that analogous to Newton's first law for translational motion, an object that is rotating will continue to do so at the same rate unless acted upon by resultant moment (in which case it would begin to accelerate or decelerate its rotational motion)]	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-53] Describe qualitatively motion in a circular path due to a centripetal force	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-54] identify the sources of centripetal force in real life examples [e.g. tension in a string for a stone being swirled around, gravity for the Moon orbiting the Earth]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-55] Illustrate that forces may produce a change in size and shape of an object	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-56] Define and calculate the spring constant [Apply the equation, Spring Constant = force/extension $k = F/x$ to solve problem involving simple Springs]	Summative	Knowledge + Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-57] Sketch, plot and interpret load—extension graphs for an elastic solid and describe the associated experimental procedures	Summative for PBA (Practical Based Assessment)	Understanding+ Application	Lab work-Question(s) will be asked in PBA
[SLO: P-09-B-58] Define and use the term ‘limit of proportionality’ for a load—extension	Summative	Knowledge+	Lab work-Question(s) will

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
graph [Including identifying this point on the graph (an understanding of the elastic limit is not required)]	for PBA (Practical Based Assessment)	Understanding + Application	be asked in PBA
[SLO: P-09-B-59] Illustrate the applications of Hooke's Law [such as that it is the fundamental principle behind engineering many measurement instruments such as the spring scale, the Galvanometer and the balance wheel of the mechanical clock]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-60] Define work done.	Summative	Knowledge	Question(s) will be asked in final examination.
[SLO: P-09-B-61] Use the equation work done = force x distance moved in the direction of the force $W = F \times d$ to solve problems	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-62] Define energy as the ability to do work	Summative	Knowledge	Question(s) will be asked in final examination.
[SLO: P-09-B-63] Explain that energy may be stored [Such as gravitational potential, chemical, elastic (strain), nuclear, electrostatic and internal (thermal) energies]	Summative	Knowledge+ Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-64] Prove that Kinetic Energy $E_k = \frac{1}{2}mv^2$ [use of equations of motion not needed; proof through kinematic graphs will suffice]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-65] Prove and use the formula for Gravitational potential energy	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-66] Use the formulas for kinetic and Gravitational potential energy to solve problems involving simple energy conversions [Make use of the conversion of energy from one form to the other, including cases involving loss of energy to the surroundings]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-67] Describe how energy is transferred and stored during events and processes [e.g. work done during transfer by mechanical work done, electrical work done and heat]	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-68] State and apply the principle of the conservation of energy	Summative	Knowledge + Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-69] Justify why perpetual energy machines do not work	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-70] Differentiate between and list renewable and non-renewable energy	Summative	Knowledge +	Question(s) will be asked in

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
sources		Understanding	final examination.
[SLO: P-09-B-71] Describe how useful energy may be obtained from natural resources [including the cases of (chemical energy stored in fossil fuels, chemical energy stored in biofuels, hydroelectric resources, solar radiation, nuclear fuel, geothermal resources, wind, tides, waves in the seas [while including references to a boiler, turbine and generator where they are used]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-72] Describe advantages and disadvantages of methods of energy generation [Limited to whether it is renewable, when and whether it is available, and its impact on the environment]	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-73] Define and calculate power [as work done per unit time and also as energy transferred per unit time. This also include applying the equations: (a) power = work done/time taken, $P = W/t$. Power = energy transferred/time taken, $P = W/t$ to solve simple problems]	Summative	Knowledge + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-74] Define and calculate efficiency [Including: (a) percentage efficiency = (useful energy output) / (Total energy input) (x100%) (b) percentage efficiency = (useful power output) / (Total power input) (x100%)]	Summative	Knowledge + understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-75] Apply the concept of efficiency to simple problems involving energy transfer	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-76] State that a system cannot have an efficiency of 100% due to unavoidable energy losses	Summative	Knowledge	Question(s) will be asked in final examination.
[SLO: P-09-B-77] Define and calculate pressure [As force per unit area. Use equation pressure = force/area $P = F/A$ to solve simple problem]	Summative	Knowledge + Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-78] Describe how pressure varies with force and area in the context of everyday examples	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-79] Analyze in situations how pressure at surface produces a force in a direction at right angles to the surface [Can make reference to experiments to verify this principle]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-80] Justify that the atmosphere exerts a pressure.	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-81] Describe that atmospheric pressure decreases with the increase in height above the Earth's surface.	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-B-82] Explain that changes in atmospheric pressure in a region may indicate a , change in the weather.	Summative	Understanding	Question(s) will be asked in final examination.

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
[SLO: P-09-B-83] Analyze the workings and applications of a liquid barometer	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-84] Justify why and analyze quantitatively how pressure varies with depth in a liquid.	Summative	Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-B-85] Analyze the workings and applications of a manometer	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-B-86] Define and apply Pascal's law [Apply Pascal's law to systems such as the transmission of pressure in hydraulic system with particular reference to hydraulic press and Hydraulic brakes on vehicles.]	Summative	Knowledge + Understanding + Application	Question(s) will be asked in final examination.
Domain C: Heat and Thermodynamics			
[SLO: P-09-C-01] Define and calculate density	Summative	Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-C-02] Justify and illustrate how to determine the density of a substance [Including for a liquid of regular shaped solid and of an irregularly shaped solids which sinks in a liquid (volume by a displacement), including appropriate calculations]	Summative	Understanding + Application	Question(s) will be asked in final examination.
[SLO: P-09-C-03] Describe, qualitatively, the particle structure of solids, liquids and gasses. [including relating their properties to the forces and distance between particles and to the motion of the particles (atoms, molecules, ions and electrons)]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-C-04] Describe plasma as a fourth state of matter [in which a significant portion of the material is made up of ions or electrons e.g. in stars neon lights and lighting streamers]	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-C-05] Describe the relationship between the motion of particles and temperature [Including the idea that there is lowest possible temperature (approximately -273 °C), known as absolute zero, where the particles have least kinetic energy]	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-C-06] State that an increase in the temperature of an object increases its internal energy	Summative	Knowledge + understand	Question(s) will be asked in final examination.
[SLO: P-09-C-07] Explain, with examples, how a physical property which varies with temperature may be used for the measurement of temperature	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-C-08] Justify the need for fixed points in the calibration of thermometers [Including what is meant by the ice point and steam point.]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-C-09] illustrate what is meant by the sensitivity, range and linearity of	Summative	Knowledge +	Question(s) will be asked in

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
thermometers.		Understanding+ Application	final examination.
[SLO: P-09-C-10] Differentiate between the structure and function of liquid-in-glass and of thermocouple thermometers	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-C-11] Analyze how the structure of a liquid in-glass thermometer affects its sensitivity, range and linearity	Summative	Application	Question(s) will be asked in final examination.
Domain E: Electricity and Magnetism			
[SLO: P-09-E-01] Describe the forces between magnetic poles and between magnets and magnetic materials [including the use of the terms north pole (N pole), South Pole (S pole), attraction and repulsion, magnetized and un-magnetized]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-E-02] Describe induced magnetism	Summative	Knowledge	Question(s) will be asked in final examination.
[SLO: P-09-E-03] State the difference between magnetic and non- magnetic materials	Summative	Knowledge	Question(s) will be asked in final examination.
[SLO: P-09-E-04] Differentiate Between Temporary and Permanent Magnets	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-E-05] Describe magnetic fields [As a region in which a magnetic pole experiences a force]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-E-06] Illustrate the plotting of magnetic field lines with a compass or iron filings	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-E-07] Draw the pattern and direction of the magnetic field lines around a bar magnet	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-E-08] State that the direction of the magnetic field at a point is the direction of the force on the N pole of a magnet at that point	Summative	Knowledge	Question(s) will be asked in final examination.
[SLO: P-09-E-09] state that the relative strength of a magnetic field is represented by the spacing of the magnetic field lines	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-E-10] Describe uses of permanent magnets and electromagnets	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-E-11] Explain qualitatively in terms of the domain theory of magnetism how materials can be magnetized and demagnetized [Stroking method, heating, orienting in north-south direction and striking, use of a solenoid]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
[SLO: P-09-E-12] Differentiate between ferromagnetic, paramagnetic and diamagnetic materials [by making reference to the domain theory of magnetism and effect of external magnetic fields on these materials]	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-E-13] Describe the nature of the Earth's magnetic field [Specifically that: - is opposite to its geographical north-south orientation - protects life on the planet from cosmic radiation - allows animals that make use of biomagnetism (e.g. many birds and turtles) to navigate during migration)]	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-E-14] Analyze applications of magnets in recording technology. [and illustrate how electronic devices need to be kept safe from strong magnetic field]	Summative	Application	Question(s) will be asked in final examination.
[SLO: P-09-E-15] State that soft magnetic materials such as soft iron) can be used to provide shielding from magnetic fields	Summative	Knowledge+ understanding	Question(s) will be asked in final examination.
Domain F: Modern Physics			
[SLO: P-09-F-01] Define and calculate average orbital speed [from the equation $v = 2\pi r/T$ Where r is the average radius of the orbit and T is the orbital period; apply this equation to solve numerical problems]	Summative	Knowledge + Understanding+ Application	Question(s) will be asked in final examination.
[SLO: P-09-F-02] Interpret and compare given planetary data [about orbital distance, orbital period, density, surface temperature and uniform gravitational field strength at the planet's surface]	Summative	Knowledge + Understanding+ Application	Question(s) will be asked in final examination.
Domain G: Nature of Science			
[SLO: P-09-G-01] Describe physics as the study of matter, energy, space, time and their mutual connections and interactions	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-G-02] Explain with examples that physics has many sub- fields, and in today's world involves interdisciplinary fields. (Students should be able to distinguish in terms of the broad subject matter that is studied between the fields: Biophysics, Astronomy, Astrophysics, Cosmology, Thermal Physics, Optics, Classical Mechanics, Quantum Mechanics, Relativistic Mechanics, Nuclear Physics, Particle Physics, Electromagnetism, Acoustics, Computational Physics, Geophysics, Climate Physics)	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.
[SLO: P-09-G-03] Explain with examples how Physics is a subset of the Physical Sciences and of the natural sciences	Summative	Understanding	Question(s) will be asked in final examination.

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks
[SLO: P-09-G-04] State that scientists who specialize in the research of physics are called Physicists	Summative	Knowledge	Question(s) will be asked in final examination.
[SLO: P-09-G-05] Brief with examples that science is a collaborative field that requires interdisciplinary researchers working together to share knowledge and critique ideas	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-G-06] Understand the terms 'hypothesis', theory' and 'law' in the context of research in the physics	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-G-07] Explain, with examples in Physics, falsifiability as the idea that a theory is scientific only if it makes assertions that can be disproven	Summative	Understanding	Question(s) will be asked in final examination.
[SLO: P-09-G-08] Differentiate the terms 'science', 'technology' and 'engineering' with suitable examples	Summative	Application	Question(s) will be asked in final examination.
Domain H: Experimentation Skills			
[SLO: P-09-10-N-01] explain, with examples, how hazards in a science lab can be classified into: ((i) physical hazards, (ii) chemical hazards, (iii) biological hazards, (v) safety hazards)	Formative for PBA	Understanding	Question will not be asked in final examination, however, it will be part of Lab work.
[SLO: P-09-10-N-02] Identify for a given experimental procedure -what would be the most appropriate personal protective equipment to wear before setting up the apparatus.	Formative for PBA	Understanding	Question will not be asked in final examination, however, it will be part of Lab work.
[SLO: P-09-10-N-03] Identify the meaning of common hazard signs in the laboratory	Formative for PBA	Knowledge	Question will not be asked in final examination, however, it will be part of Lab work.
[SLO: P-09-10-N-04] call emergency services in case of an accident in the lab	Formative for PBA	Understanding	Question will not be asked in final examination, however, it will be part of Lab work.
[SLO: P-09-10-N-05] Define and use the below terms: -True value: the value that would be obtained in an ideal measurement - Measurement error: the difference between a measured value and the true value of a quantity -Accuracy: a measurement result is described as accurate if it is close to the true value -Precision: how close the measured values of a quantity are to each other -Repeatability: a measurement is repeatable if the same or similar result is obtained when the measurement is repeated under the same conditions, using the same method, within the same experiment -Reproducibility: a measurement is reproducible if the same or similar result is obtained when the measurement is made under either different conditions or by a	Formative for PBA	Application	Question will not be asked in final examination, however, it will be part of Lab work.